

RM Assignment 2, Group 8

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Abstract

This report presents a comprehensive analysis of corporate bond pricing and portfolio credit risk management. First, we utilize a bootstrapped risk-free curve to calculate the Z-spreads and default intensities both average and piecewise constant for two defaultable bonds, further evaluating a severe credit deterioration scenario. Next, we extract real-world default probabilities from a rating transition matrix, contrasting them with market-implied metrics to highlight the impact of risk premiums during market stress. Finally, we price a portfolio of 100 investment-grade zero-coupon bonds and implement a Monte Carlo simulation based on the CreditMetrics Single-Factor Model to estimate the portfolio's 1-year 99.9% Value at Risk (VaR) and Expected Loss.

Question 1: Z-Spread Calculation

The Z-spread is defined as the basis point spread added to the risk-free yield curve such that the present value of the bond's cash flows equals its market dirty price. By hypothesis, z is constant. Applying numerical optimization via `fsolve`, we obtain:

Bond Maturity	Z-Spread
1 Year	1.4828%
2 Years	1.5232%

Table 1: Calculated Z-spreads for Bond 1 and Bond 2.

Since we see the z-spread as a "premium" on the zero rate for the fact that we are investing in something that is not completely risk-free, at least from a credit point of view as we will see in point 6, the results do not surprise us: a 2y z-spread greater than the 1y one is what we expected from the market. This result we think is given by a key factor, which is the exposition for a longer time (2y vs 1y) to the risk that the counterparty can default during the period in which we are investing in it.

Question 2: Credit Intensity and Default Probabilities

The average intensities and cumulative default probabilities ($PD = 1 - P(t, T)$) are reported below:

Metric	1 Year	2 Years
Average Intensity (λ)	2.4328%	2.4699%
Default Probability (PD)	2.4035%	4.8197%

Table 2: Intensity and Default Probability estimates.

The intensities for the two bonds are not equal, this is what we expected also by Q1. The difference between the intensities underlies a default risk which increases during time. Consequently, the 2-year PD is strictly higher not only because of the longer time horizon, but also due to this higher average intensity.

Question 3: Piece-wise Constant Intensity

In this section, we relax the assumption of a globally constant intensity and assume instead that intensity is piece-wise constant (yearly constant). The intensity for the first year (λ_1) is derived from the 1-year bond, while the intensity for the second year (λ_2) is solved using the 2-year bond price, keeping λ_1 fixed for the first year of its life.

Numerical Results

The calibrated piece-wise intensities and the resulting cumulative default probabilities are:

- **Intensity ($0 \rightarrow 1y$):** 2.4328%
- **Intensity ($1y \rightarrow 2y$):** 2.5104%
- **1y Default Probability:** 2.4035%
- **2y Default Probability:** 4.8231%

The piece-wise constant approach improves the approximation of creditworthiness of Beta also by a strictly logical point of view: previously, when we were dealing with an intensity for the whole 2 years we had an intensity which was equal from today to 1, 1.5, 2 years, while now with the piecewise constant approach we have at least an intensity a little bit more precise, since it changes after 1 year. One way to improve more the results could be to add more intensities reducing the time in which each intensity holds.

Question 4: Scenario 1 - Creditworthiness Shock

We consider a stress scenario where the market expectation of Beta's medium-term survival drops. The dirty price of the 2-year bond falls from €103.00 to €97.00, while the 1-year bond remains at €100.00. We perform a new bootstrap of the piece-wise intensity to observe the impact.

Metric	Baseline (Q3)	Scenario 1 (Shock)
1y Default Probability	2.4035%	2.4035%
2y Default Probability	4.8231%	14.7923%

Table 3: Impact of the price shock on cumulative default probabilities.

The drastic increase in the 2-year PD (from $\sim 4.8\%$ to $\sim 14.8\%$) reflects a significant widening of the credit spread in the second year to compensate for the capital loss in the bond price.

Question 5: Real-World vs. Market Probabilities

We now compare the risk-neutral probabilities derived from market prices (Scenario 1) with "real-world" (historical) probabilities derived from a rating transition matrix.

Transition Matrix Analysis

The 2-year transition matrix is calculated as M^2 . The real-world default probabilities are:

- **1y Real-World PD:** 2.00%
- **2y Real-World PD:** 4.71%
- **Conditional Real-World PD ($1y \rightarrow 2y$):** 2.7653%

Comparison with Scenario 1

The conditional probability of default in the second year under Scenario 1 is calculated as:

$$PD_{implied}(1y \rightarrow 2y) = \frac{PD_{2y} - PD_{1y}}{1 - PD_{1y}} = 12.6940\% \quad (1)$$

The implied (risk-neutral) conditional probability is significantly higher than the real-world historical probability. Our group justifies this huge discrepancy by underlying the fact that the historical data do not feel the possible volatility implied by the market of today as they are studying only what has happened in the past. One other thing that could explain this huge difference in the PDs is the shock that is imposed in Scenario 1.

Question 6: Single factor model simulations

Q6 point a:

The point a) requested to evaluate the value MtM of a ptf made up of 100 zero-coupon bonds with 2y maturity that pay €1Mln at maturity, issued by different corporations with initial rating Investment Grade. To do so, we used the following formula:

$$Ptf_{MtM} = numberissuers * N * \bar{B}^{IG}(0, t_2) \quad (2)$$

where N is the notional and $\bar{B}^{IG}(0, t_2)$ is the price of a two years IG zero coupon bond. The **ptf value is equal to** 90,131,690.79€.

Q6 point b: In order to derive the 1y forward price of a zero-coupon bond with one year residual maturity in the two different cases requested we used the formulas that were used also for the computation of $\bar{B}^{IG}(0, t_2)$, in particular we computed $\bar{B}^{IG}(t_1, t_2)$ and $\bar{B}^{HY}(t_1, t_2)$, for example for $\bar{B}^{IG}(t_1, t_2)$ we used the formula:

$$\bar{B}^{IG}(t_1, t_2) = (1 - \pi_{1,3})B(t_1, t_2) + \pi_{1,3}B(t_1, t_2/2)\eta \quad (3)$$

In the following table the prices (per 1€ notional) requested: These prices are right from a default

1y Forward	Price
IG	0.9527€
HY	0.9356€

point of view. In fact, we expect that a company with a lower rating (HY) will have an higher risk of default than the one with rating IG, so its bond will cost less than the one of the company with higher rating due to decrease in credit quality. So the result that we obtained is coherent with what we expected from the theory.

Q6 points c and d: The following results are obtained following the instructions given in the requests of the Assignment:

- Average defaults: 2
- Average downgrades: 25

We report also the marginal probabilities (which do not depend on ρ since they are marginal, so they are equal for all values of ρ):

- Probability of default: 2%
- Probability of downgrade: 25%
- Probability of status quo: 73%
- Probability of upgrade: 0% (it is not possible to have an upgrade from *IG* since it is the highest grade)

<i>Only Default</i>	<i>Expected Loss (EL)</i>	<i>VaR_{99%}</i>
$\rho = 0$	-412,357.60€	1,685,209.44€
$\rho = 0.25$	-412,035.21€	6,927,315.88€
$\rho = 0.5$	-412,490.22€	13,742,054.26€

Table 4: EL and VaR in the only defaults case

<i>Default + Downgrades</i>	<i>Expected Loss (EL)</i>	<i>VaR_{99%}</i>
$\rho = 0$	-689.93€	2,080,377.25€
$\rho = 0.25$	-462.91€	7,898,770.09€
$\rho = 0.5$	-871.10€	14,762,904.45€

Table 5: EL and VaR in the defaults + downgrades case

These results (both the one with only defaults and the one with also the downgrades) show that the VaR increases a lot as ρ increases, this is due to the structure of the *single factor model*. In fact, a higher value of ρ implies a higher weight of the systemic market factor and a loss of value of the idiosyncratic (ϵ_i) independence for each corporation causing, in case of a shock in the market, a simultaneous loss in values in most of the corporations and possible a lot of simultaneous defaults. In order to justify instead the change in value of the *EL* in the two cases, we need to underlie the fact if we consider also the downgrades and status quo maintenance (migration risk) in the computations of the *EL* and the *VaR* we are considering also the companies that 'survive' the defaults and downgrades of the other companies and preserve the status quo. Given the fact that they survived a downgrade or default, they gain value and match the value that it is loss by the other companies. This does not happen in the case in which we consider only the defaults, in fact in those cases we are just taking into account the companies that actually default and consider the ones that do not default as they preserve the status quo. In the case of only defaults, we consider the downgrades as they preserved the status quo since if we did not consider them in the counting of the companies that did not default, we would delete a number of companies not negligible. So we decided, in the case of only default, to treat the downgrades as companies that preserve their status quo

In the following graphs the visual explanation of what we have just discussed:

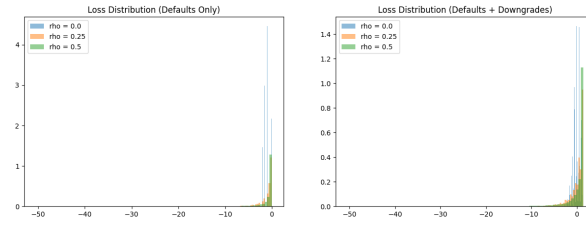


Figure 1: Loss Distribution

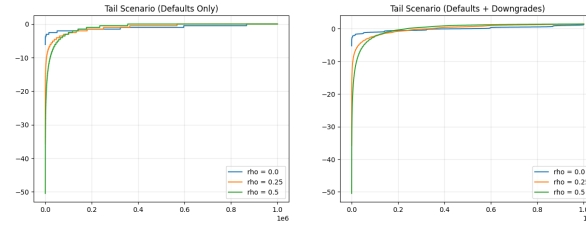


Figure 2: Tail Scenario

Mistakes in the code:

Here is a list of the mistakes that we found in the code:

MAIN (Notebook): Portfolio parameters:

- The coupon rates were both in percentage, but it is easier to use them in decimal mode.
- The expiries were calculated starting from today instead of starting from the settlement date

Fsolve function in the notebook

- In the parameters given to this function every time we used it in the code, there was written 'Today' as issue date, but in reality the issue date is the settlement date.

Q5:

- When making the product between two matrices is wrong *, we have to use @ in order to make the row*column product.

ex2utilities:

- Error: tried to import business_date_offset function from utilities.ex1 but it was actually in ex0_utilities.